

Deep Neural Networks for Detecting Illegal Fishing Vessels from Satellite Imagery

SDG 14 – Life Below Water

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1. Introduction

Illegal, Unreported, and Unregulated (IUU) fishing is a major threat to marine ecosystems and the global fishing industry. Illegal fishing depletes fish populations, damages marine biodiversity, and affects the livelihoods of coastal communities. Traditional monitoring methods such as patrol ships and manual surveillance are expensive, time-consuming, and unable to cover vast ocean regions.

Deep Neural Networks (DNNs), particularly **Convolutional Neural Networks (CNNs)**, have emerged as powerful tools for automatically detecting illegal fishing vessels from satellite imagery. Unlike shallow neural networks, deep neural networks learn multiple levels of visual features, enabling them to recognize vessels accurately even under varying weather conditions, lighting, and sea states. This supports **United Nations Sustainable Development Goal (SDG 14 – Life Below Water)** by protecting marine resources and promoting sustainable fisheries.

2. Problem Understanding

Problem Statement

Illegal fishing vessels often operate in remote ocean areas where continuous human monitoring is difficult. Manual inspection of satellite images is slow and prone to errors.

The objective is to develop a Deep Neural Network capable of:

- Detecting fishing vessels from satellite images.
- Distinguishing legal and illegal fishing activities.
- Identifying suspicious vessel movement patterns.
- Supporting maritime authorities with real-time monitoring.
- Protecting marine biodiversity and fisheries.

3. Why Deep Neural Networks Instead of Shallow Networks?

A **Shallow Neural Network** consists of only one or two hidden layers. It can learn only simple patterns such as basic shapes or pixel intensities. While suitable for simple classification tasks, shallow networks struggle with complex image recognition problems like identifying ships in noisy satellite imagery.

A **Deep Neural Network** contains many hidden layers. Each layer learns increasingly complex representations of the image, allowing the network to recognize vessels despite changes in size, orientation, shadows, clouds, or ocean conditions.

Comparison of Shallow and Deep Networks

Feature	Shallow Neural Network	Deep Neural Network
Hidden Layers	1–2	Many (10–100+)
Feature Learning	Manual or limited	Automatic hierarchical learning
Image Recognition	Limited	Highly accurate
Noise Handling	Poor	Excellent
Complex Pattern Detection	Difficult	Very effective
Accuracy	Moderate	High

Deep Neural Networks are therefore more suitable for satellite image analysis.

4. Feature Hierarchy Learning

One of the greatest strengths of Deep Neural Networks is **feature hierarchy learning**, where different layers automatically learn increasingly sophisticated image features.

Layer 1 – Low-Level Features

The first hidden layers detect basic image characteristics such as:

- Edges
- Corners
- Lines
- Brightness
- Texture

These features form the foundation for higher-level learning.

Layer 2 – Mid-Level Features

The middle layers combine low-level features to identify:

- Ship boundaries
- Hull shapes
- Wakes behind vessels
- Ocean wave patterns
- Water texture

These layers begin distinguishing ships from surrounding water.

Layer 3 – High-Level Features

The deeper layers recognize complete objects and complex relationships, such as:

- Fishing vessels
- Cargo ships
- Small boats
- Illegal fishing patterns
- Vessel movement behavior
- Suspicious ship clusters

The final layer classifies whether the detected vessel is potentially involved in illegal fishing.

This hierarchical learning enables Deep Neural Networks to perform far better than shallow networks.

5. System Architecture

The proposed detection system consists of the following stages:

1. Satellite Image Acquisition
2. Image Preprocessing
3. Feature Extraction using Deep Neural Network
4. Vessel Detection
5. Classification
6. Illegal Fishing Alert Generation

6. Data Preprocessing

Before training the model, satellite images undergo preprocessing.

Image Cleaning

- Remove noise.
- Enhance image quality.

Image Resizing

Resize all satellite images to a fixed dimension (e.g., 224×224 pixels).

Normalization

Normalize pixel values between 0 and 1 to improve learning speed.

Data Augmentation

Generate additional training images through:

- Rotation
- Flipping
- Zooming
- Cropping
- Brightness adjustment

This improves model robustness and reduces overfitting.

7. Deep Learning Model

The most suitable model is a **Convolutional Neural Network (CNN)**.

Why CNN?

CNNs are specifically designed for image recognition tasks.

Advantages include:

- Automatic feature extraction.
- High accuracy.
- Robustness to image variations.
- Efficient learning of spatial patterns.
- Reduced need for manual feature engineering.
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Popular CNN architectures include:

- VGG16
- ResNet50
- EfficientNet
- InceptionNet
- MobileNet

Among these, **ResNet50** is often preferred because its residual connections allow very deep networks to be trained effectively while maintaining high accuracy.

8. Model Training

The dataset is divided into:

- Training Set – 80%
- Testing Set – 20%

Training steps:

1. Collect labeled satellite images.
2. Preprocess images.
3. Perform data augmentation.
4. Train the CNN model.
5. Validate performance.
6. Optimize hyperparameters.
7. Test on unseen satellite images.

9. Model Evaluation

The trained model is evaluated using:

- Accuracy
- Precision
- Recall
- F1-Score
- Confusion Matrix
- ROC Curve

Example performance:

- Accuracy: 96%
- Precision: 95%
- Recall: 94%
- F1-Score: 94.5%

High scores indicate reliable vessel detection.

10. Applications

Deep Neural Networks for satellite imagery can be used in:

- Illegal fishing detection.
- Maritime border security.
- Coastal surveillance.
- Ship traffic monitoring.
- Oil spill monitoring.
- Disaster response.
- Marine conservation.
- Naval defense.

11. Advantages of Deep Neural Networks

- Automatically learn hierarchical image features.
- Detect complex vessel patterns.
- High accuracy on large datasets.
- Robust to weather and lighting variations.
- Reduce manual inspection effort.
- Real-time monitoring capability.
- Better generalization than shallow networks.
- Support large-scale ocean surveillance.

12. Limitations

- Require large labeled datasets.
- High computational requirements.
- Longer training time.
- Dependence on GPU hardware.
- Performance may decrease with poor-quality satellite images.

13. SDG 14 – Life Below Water

The proposed Deep Neural Network system directly contributes to **United Nations Sustainable Development Goal 14: Life Below Water** by:

- Reducing illegal fishing activities.
- Protecting marine biodiversity.
- Conserving fish populations.
- Supporting sustainable fisheries.
- Assisting governments in enforcing maritime laws.
- Promoting responsible use of ocean resources.
- Improving long-term marine ecosystem health.

This technology enables continuous monitoring of oceans, helping preserve aquatic life for future generations.

14. Conclusion

Deep Neural Networks significantly outperform shallow neural networks in detecting illegal fishing vessels from satellite imagery because they learn **hierarchical features**, progressing from simple edges and textures to complete vessel structures and complex fishing patterns. Using CNN architectures such as **ResNet50**, the system can accurately detect suspicious vessels, support maritime surveillance, and reduce illegal fishing activities. By enabling efficient monitoring of vast ocean regions, this approach plays a vital role in achieving **SDG 14 – Life Below Water**, ensuring the conservation and sustainable use of marine resources.